Emily Tran

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Education

California State Polytechnic University, Pomona Aug 2022 – May 2026
Bachelor of Science in Computer Science, Minor in Data Science *Pomona, CA*
GPA: 3.74 | Magna Cum Laude

Projects

Gambler's Addict | *Unity, C#, Gameplay Systems, Adaptive AI*
- Designed and implemented adaptive betting, economy, and difficulty-scaling systems for a 2D bullet-hell roguelike in Unity
- Engineered a score-weighted boss AI that dynamically selected attack patterns and scaled encounter difficulty
- Built persistent progression systems including save-state management, adaptive reward multipliers, and cross-scene data persistence
- Developed debugging and gameplay visualization tools to support balancing and playtesting iteration

Funk vs Fright | *Unity, C#, Animation Systems, Gameplay Programming*
- Led a 4-person interdisciplinary team as project lead, coordinating gameplay, art, and UI development for a 2.5D rhythm-combat game in Unity
- Implemented timing-sensitive combat mechanics, animation state systems, and gameplay feedback loops
- Created UI assets, sprite animations, and visual feedback systems to improve combat readability
- Won 1st Place overall among competing game jam submissions

Calendar Meet | *React, Node.js, PostgreSQL, AWS, Github Actions*
- Collaborated on development of a full-stack scheduling platform
- Implemented AWS deployment infrastructure and automated CI/CD pipelines using GitHub Actions
- Deployed and maintained a React/Node.js/PostgreSQL application in a cloud-hosted environment

Experience

Virtual Reality Lab — Cal Poly Pomona Mar 2025 – Jun 2025
Unity Developer *Pomona, CA*
- Collaborated within a 4-person development team on a VR zombie survival FPS built in Unity using XR Interaction Toolkit
- Developed VR gameplay interaction systems, enemy AI behaviors, and spatial 3D audio systems
- Conducted playtesting, debugging, and iterative gameplay improvements for usability and immersion

Research Experience for Undergraduates (REU) — Texas A&M University May 2024 – Aug 2024
Applied Computational Robotics Researcher *College Station, TX*
- Developed Pygame simulations to evaluate autonomous rendezvous algorithms
- Implemented and analyzed A* and D* Lite pathfinding algorithms
- Conducted comparative testing and presented technical findings in a collaborative research environment

Technical Skills

Languages: C#, Python, Java, JavaScript, SQL
Game Development: Unity, Gameplay Programming, Game AI, XR Interaction Toolkit
Tools: Git, GitHub, VS Code, Aseprite
Additional: React, Node.js, PostgreSQL, AWS, GitHub Actions